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Tutorial-2

1. Write a void function which takes in two integers as parameters. The function will sum the two numbers and print the answer.

Ans:

Source Code:

```c
#include <stdio.h>

void add_2_numbers(int a,int b){
 int sum=a+b;
 printf("%d + %d = %d",a,b,sum);
}

void main(){
 int num1,num2;
 printf("Enter a number : ");
 scanf("%d",&num1);
 printf("Enter another number : ");
 scanf("%d",&num2);
 add_2_numbers(num1,num2);
 printf("\n");
}
```

Output:

```
Enter a number : 10
Enter another number : 206
10 + 206 = 216
```

2. Write a function which prints the area and perimeter of a circle. The function should only have one parameter; the radius.

Ans:

Source Code:

```c
#include<stdio.h>

void area_and_perimeter(float radius){
 float area=3.14*radius*radius;
 float perimeter=2*3.14*radius;

 printf(" Area = %.2f \n",area);
 printf(" Perimeter = %.2f \n",perimeter);
}

void main(){
 float r;

 printf("Enter the radius of the circle : ");
 scanf("%f",&r);
 area_and_perimeter(r);

 printf("\n");
}
```

Output:

```
Enter the radius of the circle : 25
 Area = 1962.50
 Perimeter = 157.00
```

3. Write a void function to convert a decimal number into a binary number. The function should take in one parameter (the decimal value (integer)). You will need to think about how to print this out, you ideally should use a character/integer array to store the result. For this question, restrict your output to 8 bits max to save time and memory.

Ans:

Source Code:

```c
#include <stdio.h>
#include <string.h>

void decimal_to_binary(int number){
 char binary_num[8];
 int remainder,i=0,num=number;

 while(number!=0){
  remainder = number%2;
  binary_num[i]=remainder+'0';
  number=number/2;
  i++;
  }
 printf("Binary equivalent of %d = ",num);

 for(int j=i-1;j>=0;j--){
 printf("%c",binary_num[j]);
 }
 }

int main(){
 int num;

 printf("Enter an integer : ");
 scanf("%d",&num);
 decimal_to_binary(num);

 printf("\n");
}
```

Output:

```
Enter an integer : 50
Binary equivalent of 50 = 110010
```

4. Write a function to return the sum from 1 to a value specified by the single parameter of a function. For example, function(9) should return 45 as this is 1+2+3+4+5+6+7+8+9

Ans:

Source Code:

```c
#include<stdio.h>

int sum1(int num){
 int num_to_return=0;

 for(int i=1;i<=num;i++){
  num_to_return=num_to_return+i;
 }
 return num_to_return;
}

int main(){
 int a,output;
 printf("Enter a number : ");
 scanf("%d",&a);
 output=sum1(a);
 printf("The sum of the numbers from 1 to %d is %d.",a,output);

 printf("\n");
}
```

Output:

```
Enter a number : 10
The sum of the numbers from 1 to 10 is 55.
```

5. Write a void function which finds and prints the mid-point coordinates of a line. The function should take in four parameters (x1, y1, x2 and y2).

Ans:

Source Code:

```c
#include <stdio.h>

void mid_point(float x1,float y1,float x2,float y2){
 float mid_point_x,mid_point_y;

 mid_point_x=(x1+x2)/2;
 mid_point_y=(y1+y2)/2;

 printf("Mid-point of the line : (%f,%f).",mid_point_x,mid_point_y);

}

int main(){
 float x1,y1,x2,y2;

 printf("Enter the coordinates of the first endpoint of the line with a space in the middle : ");
 scanf("%f %f",&x1,&y1);

 printf("Enter the coordinates of the second endpoint of the line with a space in the middle : ");
 scanf("%f %f",&x2,&y2);

 mid_point(x1,y1,x2,y2);

 printf("\n");
 return 0;
}
```

Output:

```
Enter the coordinates of the first endpoint of the line with a space in the
middle : 2 5
Enter the coordinates of the second endpoint of the line with a space in the
 middle : 3 9
Mid-point of the line : (2.500000,7.000000).
```

6. Write a void function which takes one integer (n) as a parameter. Your function should then print out all triangular numbers from 1 up to the nth term. You will need to research the formula to find triangular numbers.

Ans:

Source Code:

```
#include<stdio.h>

void triangular_numbers(int n){
 printf("First %d triangular numbers : \n",n);
 for(int i=1;i<=n;i++){
  int tn=(i*(i+1))/2;
  printf("%d\n",tn);
 }
}

int main(){
 int n;
 printf("Enter a number : ");
 scanf("%d",&n);
 triangular_numbers(n);

 printf("\n");
 return 0;
}
```

Output:

```
Enter a number : 10
First 10 triangular numbers :
1
3
6
10
15
21
28
36
45
55
```

7. Write a function which returns an appropriate value for the formula “F = MA.” F = Force. M = Mass. A = Acceleration. Depending on which variable is set to “NAN” when the function is called, your function should work it out and return that value.

Ans:

Source Code:

```c
#include <stdio.h>
#include <math.h>

float calculation(float mass,float acceleration,float force){
 if(isnan(mass)){
  return force/acceleration;
 }else if(isnan(acceleration)){
  return force/mass;
 }else if(isnan(force)){
  return mass*acceleration;
 }else{
  return -1;
 }
}

int main(){
 float m,a,f,result;

 while(1){
 printf("Enter mass (enter NAN for unknown value): ");
 scanf("%f",&m);

 printf("Enter accelaration (enter NAN for unknown value): ");
 scanf("%f",&a);

 printf("Enter force (enter NAN for unknown value): ");
 scanf("%f",&f);

  if(isnan(m)+isnan(a)+isnan(f)>1){
  printf("Error: More than one value is NAN.\n");
  continue;
 }

 result=calculation(m,a,f);

 if(isnan(m)){
  printf("Mass = %f .",result);
 }else if(isnan(a)){
  printf("Acceleration = %f .",result);
 }else if(isnan(f)){
  printf("Force = %f .",result);
 }else{
  printf("Error: Nothing to calculate.\n");
  continue;
 }
  break;
 }

 printf("\n");
 return 0;
}
```

Output:

```
Enter mass (enter NAN for unknown value): nan
Enter accelaration (enter NAN for unknown value): nan
Enter force (enter NAN for unknown value): nan
Error: More than one value is NAN.
Enter mass (enter NAN for unknown value): 123
Enter accelaration (enter NAN for unknown value): 132
Enter force (enter NAN for unknown value): 3545
Error: Nothing to calculate.
Enter mass (enter NAN for unknown value): 50
Enter accelaration (enter NAN for unknown value): 45
Enter force (enter NAN for unknown value): nan
Force = 2250.000000 .
```

8. Write a single function which returns and represents all SUVAT equations. You will need to do some research online on the formulas. This will test your formula rearranging skills as you can work out different variables from one equation however, all the inputs may vary so you will need implement as many formulas as possible. Your program should always be able to work out one variable when the other four variables have been set.
   S = distance
   U = initial velocity
   V = end velocity
   A = acceleration
   T = time

Ans:

Source Code:

```c
#include <stdio.h>

float suvat(int choice,float v1,float v2,float v3,float v4){
 switch(choice){
   case 1:
    return (v1*v4 + 0.5*v3*v4*v4);
   case 2:
    return (v2-v3*v4);
   case 3:
    return (v2+v3*v4);
   case 4:
    return (v3-v2)/v4;
   case 5:
    return (v3-v2)/v4;
}
}

int main(){
 float s,u,v,a,t,output;
 int choice;
  printf("\nWhat do you need to find : \n1)Distance(s)\n2)Initial Velocity(u)\n3)End Velocity(v)\n4)Acceleration(a)\n5)Time(t)\n");
  scanf("%d",&choice);
  switch(choice){
   case 1:
    printf("\nEnter initial velocity : ");
    scanf("%f", &u);
    printf("\nEnter end velocity: ");
    scanf("%f", &v);
    printf("\nEnter acceleration: ");
    scanf("%f", &a);
    printf("\nEnter time: ");
    scanf("%f", &t);
    output=suvat(choice,u,v,a,t);
    printf("\nDistance : %.2f.",output);
    break;
   case 2:
    printf("\nEnter distance : ");
    scanf("%f", &s);
    printf("\nEnter end velocity: ");
    scanf("%f", &v);
    printf("\nEnter acceleration: ");
    scanf("%f", &a);
    printf("\nEnter time: ");
    scanf("%f", &t);
    output=suvat(choice,s,v,a,t);
    printf("\nInitial velocity : %.2f",output);
    break;
   case 3:
    printf("\nEnter distance : ");
    scanf("%f", &s);
    printf("\nEnter initial velocity: ");
    scanf("%f", &u);
    printf("\nEnter acceleration: ");
    scanf("%f", &a);
    printf("\nEnter time: ");
    scanf("%f", &t);
    output=suvat(choice,s,u,a,t);
    printf("\nEnd velocity : %.2f",output);
```

```
    break;
   case 4:
    printf("\nEnter distance : ");
    scanf("%f",&s);
    printf("\nEnter initial velocity : ");
    scanf("%f", &u);
    printf("\nEnter end velocity: ");
    scanf("%f", &v);
    printf("\nEnter time: ");
    scanf("%f", &t);
    output=suvat(choice,s,u,v,t);
    printf("\nAcceleration : %.2f",output);
    break;
   case 5:
    printf("\nEnter distance : ");
    scanf("%f",&s);
    printf("\nEnter initial velocity : ");
    scanf("%f", &u);
    printf("\nEnter end velocity: ");
    scanf("%f", &v);
    printf("\nEnter acceleration: ");
    scanf("%f", &a);
    output=suvat(choice,s,u,v,a);
    printf("\nTime : %.2f",output);
    break;
   default:
    printf("\nInvalid input.");
  }
 printf("\n");
 return 0;
}
```

Output:

```
What do you need to find :
1)Distance(s)
2)Initial Velocity(u)
3)End Velocity(v)
4)Acceleration(a)
5)Time(t)
1

Enter initial velocity : 23

Enter end velocity: 50

Enter acceleration: 2

Enter time: 6

Distance : 174.00.
```

```
What do you need to find :
1)Distance(s)
2)Initial Velocity(u)
3)End Velocity(v)
4)Acceleration(a)
5)Time(t)
2

Enter distance : 340

Enter end velocity: 100

Enter acceleration: 5

Enter time: 4

Initial velocity : 80.00
```

```
What do you need to find :
1)Distance(s)
2)Initial Velocity(u)
3)End Velocity(v)
4)Acceleration(a)
5)Time(t)
3

Enter distance : 40

Enter initial velocity: 34

Enter acceleration: 6

Enter time: 40

End velocity : 274.00
```

```
What do you need to find :
1)Distance(s)
2)Initial Velocity(u)
3)End Velocity(v)
4)Acceleration(a)
5)Time(t)
4

Enter distance : 50

Enter initial velocity : 50

Enter end velocity: 200

Enter time: 14

Acceleration : 10.71
```

```
What do you need to find :
1)Distance(s)
2)Initial Velocity(u)
3)End Velocity(v)
4)Acceleration(a)
5)Time(t)
5

Enter distance : 1000

Enter initial velocity : 45

Enter end velocity: 79

Enter acceleration: 4

Time : 8.50
```

```
What do you need to find :
1)Distance(s)
2)Initial Velocity(u)
3)End Velocity(v)
4)Acceleration(a)
5)Time(t)
6

Invalid input.
```

9. Write a void function which solves simultaneous equations. Your program will take six parameters. E.g. function(double a, double b, double c, double d, double e, double f){} By solving simultaneous equations, you are finding where the two lines cross each other, so your function should print an x and y coordinate.
   a = number in front of x of equation one b = number in front of y of equation one c = constant of equation one d = number in front of x of equation two e = number in front of y of equation two f = constant of equation two

Ans:

Source Code:

```c
#include <stdio.h>

void equations(double a,double b,double c,double d,double e,double f){
 double determinant=(a*e)-(b*d);
 if(determinant==0){
  printf("No unique solution(lines are parallel or identical)\n");
 }else{
  double x=(c*e-b*f)/determinant;
  double y=(a*f-c*d)/determinant;

  printf("X = %lf\n",x);
  printf("Y = %lf",y);
 }
}

int main(){
 double a,b,c,d,e,f;

 printf("Enter value of a for the equation(ax+by=c) : ");
 scanf("%lf",&a);
 printf("Enter value of b for the equation(ax+by=c) : ");
 scanf("%lf",&b);
 printf("Enter value of c for the equation(ax+by=c) : ");
 scanf("%lf",&c);
 printf("Enter value of d for the equation(dx+ey=f) : ");
 scanf("%lf",&d);
 printf("Enter value of e for the equation(dx+ey=f) : ");
 scanf("%lf",&e);
 printf("Enter value of f for the equation(dx+ey=f) : ");
 scanf("%lf",&f);

 equations(a,b,c,d,e,f);

 printf("\n");
 return 0;
}
```

Output:

```
Enter value of a for the equation(ax+by=c) : 124
Enter value of b for the equation(ax+by=c) : 523
Enter value of c for the equation(ax+by=c) : 46346
Enter value of d for the equation(dx+ey=f) : 1435
Enter value of e for the equation(dx+ey=f) : 53351
Enter value of f for the equation(dx+ey=f) : 235321
X = 400.601015
Y = -6.364294
```

```
Enter value of a for the equation(ax+by=c) : 2
Enter value of b for the equation(ax+by=c) : 4
Enter value of c for the equation(ax+by=c) : 10
Enter value of d for the equation(dx+ey=f) : 1
Enter value of e for the equation(dx+ey=f) : 2
Enter value of f for the equation(dx+ey=f) : 5
No unique solution(lines are parallel or identical)
```